

UNIVERSITY PARTNER



Milestone 4 Professionalism & Early Prototype

“EDIT EASE- A Video Management Platform”

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Introduction

EditEase is a video management and retrieval platform. It's built for editors and content teams who work with massive mountains of raw footage. The main idea is to take unstructured video files and turn them into structured, searchable project assets by creating indexes for each scene with timestamps, thumbnails and descriptive metadata. EditEase may even introduce AI insufficiency signs in the future - think of "confidence"-based scene cues and primitive emotion hints to the user, to aid the browsing process and to reduce the amount of manual scrubbing time.

Because EditEase is applicable to real world video, it's likely to be processing personal data such as faces, voices, clothes, locations, or any other identifiable details. This raises a bunch of responsibilities of professionalism that go beyond the tech getting it to work. Even a perfectly built system can do harm, should it be pushed out without safeguards, implement in the wrong context, or when people take AI outputs for granted as fact. That's why this report takes a look at EditEase from 4 required professionalism angles - Social, Ethical, Legal, and Security and for each of these angles.

1. Social Aspect

Stakeholders and real world impact

This affects multiple stakeholders including editors, clients/organizations, filmed subjects, and developers/maintainers.

- * Editors and content teams - the key users who will get most benefit out of faster searching, more fluent clip selection and consistent organisational.
- * Organisations/clients - secondary users who have expectations of responsible working of footage and how it will work through.
- * Film subjects - the individuals who may be shown in the film; misuse could be harmful to them if the content is given to the wrong people or misinterpreted.

- * System maintainers/admins - bears all the responsibility of configuration, access control and deployment choices.

The fundamental social effect is that it alters the way that teams handle raw material. Instead of 'watch everything', the workflow is then changed to 'search, filter and jump to scenes'. That increase in efficiency can also lead to concentration of power: whoever has control of tags, metadata or user permissions could potentially influence what people see or miss. So the tool has to be developed in a way that reduces misuse of it and creates no feeding of biased results (ICO, 2024).

1.1 Social Pros (benefits in terms of users and teams)

Pros:

- * Time saving and less fatigue - timelines are often scrubbed by the editors for endless hours. Scene-Level Indexing eliminates time-consuming repetitive viewing and optimizes the delivery process, reducing cognitive load, and enhancing job satisfaction.

- * Improved team coordination - shared tags and structured metadata make it easier for teams to divide work, hand off projects and ensure consistency between editors.

- * Better project continuity - for long running projects a searchable index serves as a kind of "shared memory" that new team members can use to catch up quickly.

- * Safer operation when governed by governance - if the system is implemented with clear rules about high-risk use of AI, and protecting data, it may actually lead to safer operational habits - such as humans taking the time to check back automated hints for accuracy, thereby encouraging caution and correct interpretation (ICO, 2024).

1.2.Social Cons (negative impact or likely to cause harm)

Cons:

- Repurposing for surveillance - A tool intended for editing could easily be repurposed as a surveillance tool for employees or to track their behaviour if boundaries aren't set. That end use may bring about social harm however even if the goal set forth was efficiency at the onset (ICO, 2024).
- Misinterpretation of AI hints - if emotion cues or "scene meaning" labels are interpreted as objective truth, this can hurt people's reputations or careers. The ICO identifies inferences about individuals to be high risk and must be dealt with carefully (ICO, 2024).
- Access inequality and usability gaps - if the interface is too technical or the filters opaque, then non-technical team members may end up being dependent on a handful of 'power users' and create internal power imbalances;
- Bias from over- filtering - Having a high dependence on tags risks creating a myopic picture of the footage, perpetuating one narrative (e.g. selecting only dramatic parts of footage over and over again because they're tagged as being high intensity).

2. Privacy and Data Handling Principles

EditEase works with personal data indirectly since the videos could contain faces, voices, and recognizable locations. The system still contacts delicate data even in case the primary task is the clip organization. Best data handling practice implies adherence to such principles as:

2.2 Data Minimization

The system is designed to store primarily metadata and indexes and not complete video copies. Scene-level attributes that are maintained in the database include:

- start and end timestamps
- duration
- thumbnail path
- estimated type of scene and confidence.
- emotion history and prevailing emotion (where appropriate)
- fields which are only reviewed such as status and notes.

This maintains low chances of overloading data. The real video data pursues across the dataset folder instead of actively being drawn to MongoDB.

2.3 Purpose Limitation

Any data that is extracted is purposely obtained to meet its one purpose to search, filter and arrange clips. It is not applied in profiling people and drawing conclusions about them. The outputs of emotion and face recognition are helper features on workflow editing.

2.4 Retention Awareness

Videos are typically project-based in the real production setting. Metadata and thumbnails used in a project could become unnecessary once it is completed. A professional system must have the potential of deletion or archiving. the prototype is purely functional, whereas we are aware of the principle of retention, and we intend to introduce cleanup facilities in the future.

2.5 Controlled Exposure

Currently the prototype is local and operating with:

- Flask API on 127.0.0.1:5000
- Streamlit UI locally

This reduces the chances of being exposed to accidents as published endpoints are in the online system. In case it can be deployed later, we will certainly be able to add access controls and authentication.

3. Ethical AI Considerations: Bias, Accuracy, and Interpretability

EditEase relies on pretrained models and rule-based analysis to classify and detect emotions, introducing the common AI issues of bias, reliability, and transparency.

3.1 Emotion Recognition Limitations

Facial expressions are not perfect in determining emotions. Accuracy hinges on:

- visualization of the face (range, blurriness, obstructions)
- face orientation (upside -down, rotated, side-profile)
- lighting and video quality
- differences of cultural expression.

We noticed numerous happy tags on the scenes where there was no visible human face, such as product shots, during testing. That is an issue, since the raw materials of an inaccurate output may undermine faith and become abused.

Mitigation implemented:

Face-guard: in case no face is present in the thumbnail or sampled frames, the system assigns the system emotion output to null. That prevents irrelevant scenes to which false feelings can be affixed.

3.2 Bias Considerations

Off-the-shelf models are subject to skewed distribution in demographics- skin tone, age, face structure, cultural expression. Because EditEase can be applied to the Nepali and a wide range of South Asian footage, there is a threat of prejudice. The prototype thus does not give emotion results as being definitive. The UI indicates confidence scores and raw metadata to positively influence reviewers to recheck.

3.3 Human-in-the-Loop Review

In order to maintain prudent utilization, EditEase has implemented a manual review process:

- every scene receives a reviewed flag of a boolean.
- the user may disregard the scene names and emotions in the interface.
- a notes field is optional and allows context to be added by the editors.

This forms a ground-truth labeling which can be utilized in later training and testing models.

3.4 Ethical Pros (responsible design strengths)

Pros:

1. Privacy-aware engineering direction: A responsible system aims to minimise exposure: store only what is required for the workflow and avoid unnecessary duplication or distribution of raw footage. This aligns with responsible data handling expectations (ICO, 2024).
2. Human-in-the-loop decision model: EditEase is ethically safer when automated outputs do not make final decisions and users remain accountable for interpretation and usage (ICO, 2024).
3. User control and correction: Ethically, users should be able to override labels, remove misleading tags, and add notes that provide context. This prevents the AI output from becoming permanent “truth.” This matches the accountability emphasis in AI and data protection guidance (ICO, 2024).
4. Purpose limitation mindset: Ethical use means the tool supports editing/retrieval, not secondary uses like profiling or monitoring. Defining clear boundaries is ethically important (ICO, 2024).

3.5 Ethical Cons (risks and moral concerns)

Cons:

- Consent issues: The individuals being recorded may have no idea that their footage will be converted to searchable indexes and analysed based on other emotional issues. Even lawfully acquired footage must be cautiously dealt with and use restrictions (ICO, 2024).
- Sensitivity of emotion inference: Sensing emotions is a difficult task that is quite context reliant. Suggestive indicators of emotion (without clear warnings) may confuse users into incorrect assumptions of the intent or trustworthiness (ICO, 2024).
- Equity and partiality: AI may be unsuccessful on the basis of the lighting, camera angles, skin colors, and cultural manifestations. Design must not claim too much, must provide safe defaults and must press to validate (ICO, 2024).
- Over-reliance / automation bias: The users may have faith in the system since it is automated. It becomes a particularly high risk in cases when less experienced editors apply it on the clock (ICO, 2024).

4. Legal and Regulatory Considerations

Even as a prototype, a professional artefact must acknowledge legal constraints.

4.1 Copyright and Ownership of Footage

A majority of video footage belongs to the creator or the client and is under copyright. The thumbnails obtained are derived works. Hence, the system should not be applied with hacked footage. Academically, videos are supposed to be recordings of the student or datasets permitted by him or her.

4.2 Compliance with Privacy Norms

This prototype is localized, but in case it is applied in real organizations, the adherence to the principles of privacy cannot be neglected. Without legal references, the system ought to be in line with the general privacy requirements, which involve reducing storage, limiting access and secondary use.

4.3 Tool Licensing

OpenCV, FFmpeg, MongoDB, Flask, and Streamlit are some of the popular libraries used in EditEase. An expert report recognizes the fact that every tool is licensed. These tools are used in the development and research of the prototype. Licensing and the conditions of commercial use should be verified in future productions deployment.

4.4 Legal pros:

The sensitivity to early licensing decreases future risk: Too many project boards of students encounter roadblocks due to thinking about licensing too late. Early licensing helps us to be prepared to professional work.

The clarity of Flask license: Flask is BSD -3 -Clause licensed, but it is generally permissive and does not cause much issue with our deployment as long as we observe the license conditions (Pallets/Flask, n.d.).

Streamlit license transparency: Streamlit is Apache-2.0 based with transparency. Its responsible usage would include reading those terms and attributing them where necessary (Streamlit, n.d.).

4.5 Legal Cons

Cons:

MongoDB SSPL can establish stiff commitments: should we provide EditEase as a service (e.g. in the form of a hosted service) the SSPL may place additional obligations. To deliver a SaaS in the future, we will have to re-verify compliance (MongoDB, n.d.).

busyboard of mixed licensing: The mixture of components may unintentionally become the cause of obligations when we do not follow the licenses in a single location (MongoDB, n.d.; Streamlit, n.d.; Pallets/Flask, n.d.).

Risk of distribution ambiguity: The fact that they do not specify whether EditEase is packaged software, an internal tool, or a hosted service makes it more difficult to determine compliance, particularly when dealing with SSPL instances (MongoDB, n.d.).

5. Security and Reliability Considerations

The concept of security cannot be ignored since it is possible to reveal sensitive data even with metadata (timestamps, faces in thumbnails, and the types of the scenes).

5.1 Threats and Risks

Key risks include:

- MongoDB not secured with authentication.
- API endpoints are abused and misused to query or export content.
- file locales through API responses.
- feature of export getting exploited to extract a lot of clips in a short time.
- unintentional corruption of indexes because of a mixture of irregular types of JSONs.

5.2 Current Prototype Controls

Some of the controls in this milestone include:

- local-only deployment (127.0.0.1)
- parameterized query parameters API.
- NumPy types to native types conversion to guarantee the stability of JSON.
- export ffmpeg -ss start -t duration -c copy to quickly cut a video without re-coding.

5.3 Planned Hardening (Future Work)

Some of the ways to improve in the future are:

- I have authenticated MongoDB and connected it to the localhost exclusively.
- Authenticating API key, in case deployed on network.
- checking and secure processing of file paths.
- exporting clip audit logs.
- The latter will be taken into consideration in subsequent milestones when the system will be no longer local prototype.

5.4 Pros of Security (current strengths / planned controls)

Pros:

Security governance attitude: ISO/IEC 27001 encourages a methodical risk attitude, policies and controls. Regarding security, it is a good professional gesture to take it as a continuous process rather than a one-time adjustment (ISO, 2022).

Direction of identity and authentication: Development of the login flows based on the recognized identity standards reduces the risk of weak security modes (NIST, 2025).

Risk awareness of API: The OWASP API Top 10 assists us in concentrating on the subjects of authentication, authorization, input validation, rate limits, and safeguarding errors (OWASP, 2023).

Controlled access objectives: RBAC implementation and restricted export to authorized users reduces the data exposure risk (ISO, 2022; OWASP, 2023).

5.5 Cons(security risks and weaknesses during the prototype phase)

Cons:

- Broken authentication / weak identity controls: In case the authentication of the prototype is not complete, the accounts would be compromised or gated. According to the identity guidance of NIST, we are supposed to use the following controls in production (NIST, 2025).
- broken object-level authorization: APIs that enable users to export or get access to assets should ensure that they can only view what they are permitted to view- this is a typical OWASP risk (OWASP, 2023).
- Too much metadata: APIs providing a lot of metadata or excessive focus on error information may disclose sensitive information. This is specifically noted as an API risk pattern (OWASP, 2023).
- Absence of auditing and traceability: There are no logs or audit trails, and it is difficult to check the misuse of it, which undermines accountability and governance (ISO, 2022).

- Gap in operational hardening This is implemented into production: it involves secured configurations, secrets management, and database access control, and these areas are not always implemented in prototypes (ISO, 2022).

6. Professional Engineering Practice and Prototype Development Approach

The following prototype was created following an organized design engineering method:

6.1 Modular Design

The pipeline will be divided into modules:

- detectscenes.py to detect scenes.
- thumbnail extraction frame
- thumbnail extraction frame
- emotiondetect.py to detect emotion.
- scenetypedetect.py conversion logic of the logic of scattering scenes.
- runpipeline.py to create indexes on individual videos.
- Flask API- retrieve and export.
- Streamlit views of browsing, filtering, labeling, exporting.

This allows the system to be easier to debug and to extend since it is developed in a more of a modular manner.

6.2 Robustness Improvements

There were some edge cases which were met and discussed:

- JSON serialization problem with NumPy float32 values -JSON serialization solved by converting some many numbers to NumPy using makejsonsaf.
- No cuts in certain videos (single continuous clips) - should not cut: should still produce at least one scene index; fallback: should treat the whole video as a scene.
- Non-face-based false predictions of emotions – mitigation: face detection guard sets emotion fields to zero.

7. Early Prototype of the Artefact (Implementation Evidence)

The initial prototype provides an understanding of the main idea: ingest raw videos, index scenes, search/filter, export clips.

7.1 Pipeline Flow (End-to-End)

The present workflow is the following system workflow:

1. Video inputs are put within a dataset folder (e.g., editeaseapp/data/).
2. Scene segmentation is based on cut-based logic in detection of scene boundaries, with no cuts, the entire clip is one scene.
3. The extraction of the thumbnails has chosen within each identified scene representative frames.
4. Emotion analysis is run:
5. in case faces are spotted, emotion is sampled within a scene (timeline approach Option A)
6. in case of face not found emotion is recorded as null.
7. Scene-type classification operates based on motion and face presence as heuristics to categorise scenes as testimonial, b-roll, audience reaction, establishing or otherwise.
8. Index generation will generate a scene level JSON file in each video and add records to the database in MongoDB.
9. Search/filter and export are made possible with Backend API (Flask).
10. Frontend UI (Streamlit) represents thumbnails, metadata, emotion timeline, manual labeling, and export.

7.2 Prototype Output Data

Indexed scenes have the following fields in them;

- video, sceneid
- startsec, endsec, and durationsec.
- thumbnail path
- scenelabel, sceneconfidence, scenedebug.
- emotiontimeline, predominantemotionoverall (or null)

- checked flag and optional notes.

This format has been used since it can be search and filterable and can be used with MongoDB queries. It is also helpful in subsequent ML training as manual labels are transformed into the ground truth.

8. Conclusion

Finally, Milestone 4 indicates that EditEase is no longer a crass hypothesis, it is a workable initial model that is presentable and, in fact, addresses not only the technical viability but the practical accountability as well. It is now possible to feed the raw video into the system and transform it into data that is structured and searchable without the need to scrub timelines by hand. That is what we went to do: stick thousand tons of video clips in each category where we are going to need them instead of fully editing. The workflow, albeit in its crude state, demonstrates that automated indexing (performing scene segmentation, generating thumbnails and extracting metadata) can dramatically reduce the time spent to locate the desired clips.

Professionally, this milestone also highlights the fact that an AI-assisted system is not merely about accuracy: it is also about safe responsible use. The project does not forget that sensitive personal information can be stored in the videos: faces, locations, client information. To provide a prototype that is practical, we make privacy-progressive decisions: it doesn't store entire videos in the store, rather it relies on sparse metadata, it is deployed initially to a local environment, and it promotes human review with a review-and-override format. These measures are used to ensure the system is ethically focused, particularly as, with automated predictions, such as detecting emotions, there is a chance of misinterpretation, when shown as facts. One of the most relevant professional enhancements, in this case, is our approach to the AI limits in reality. The preliminarily performed tests indicated that even pre-trained emotion models are able to confuse in the case of no visible faces, to produce misleading results, which is harmful to trust and utility. Instead of turning a blind eye to this, EditEase provides security measures such as face checks and the display of null emotion in the absence of facial evidence. This is the thought process of a responsible engineer: when a model cannot be used to support a prediction, the system is supposed to respond with unknown rather than provide false and confident sales talk. The system can be more straightforward with the null forcing of shaky emotion cases, and can be much simpler to interpret and users can more easily be able to manually check the system. The prototype further also provides a solid foundation by establishing a human-in-the-loop process of labeling in the future of AI work. Since the system labels the reviews, can add manual scene labels and optional notes, it simply establishes a system to gather ground truth. That is essential since the long-term aim is not to

permanently use pre-trained on the example of hard thresholds, but to gradually replace them with the one custom-trained at the example of real EditEase videos. Milestone 4 is not merely a working prototype because that is a strategic set-up stage wherein the system begins to construct the precise structured dataset that it will require later on in the training, assessment and subsequent enhancement.

Altogether, the Milestone 4 demonstrates that the project is developing in a proper direction, both technically and professionally. Indexing pipeline core is operational, the results are clean and database friendly and the failure cases that are caused by key failures are addressed in a responsible manner and not ignored. Above all, the system is developed on the assumption that AI is meant to assist, but not make decisions. The product is a premature artefact that can be used today to sort clips and at the same time be able to adopt safe deployment, reliable interpretation and future revisions based on learning. This milestone is a significant advance towards a scalable video-organisation system and it also defines the next steps: refinement of segmentation of continuous scenes, increasing the hand labeling of data construction, and replacing guessing with concrete and assessed machine-learning models in later milestones.

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